A common loop-topological carrier for electroweak and emergent-gravity sectors

(Technical consistency note across Papers 1–4)

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Abstract

This note performs a cross-file consistency audit across the four companion papers of the corpus (Papers 1–4). It is a *technical consistency note*, not an independent paper with new physical claims. For the corpus-wide entry point, claim grammar, corpus map, and falsification handles see the reader's-guide manifest [1]; the present bridge note operates at the lemma level (P1↔P4 cross-consistency). We document that the electroweak sector of Paper 1 [2] and Paper 3 [4] and the cosmological / Einstein sector of Paper 4 [5] *share the same finite inputs*: the same five measured causal-wave transport coefficients (α_ξ, $D(\Omega)$, β_π, γ, $\varepsilon^2_{\text{sync}}$) = (0.90082, 0.83996, 0.93791, 0.10021, 0.05000), the same finite loop-class library (Lemmas 1–8 plus a pure-sync class), and the same T-parity source axiom (defect field Ξ T-odd fixing $\theta_{\text{em}} = \pi$). The two sectors specialise differently along the spinor-trace component count n, but they overlap at $n = 1$: the QFT sector lives entirely at $n = 1$; the Einstein sector spans $n \in \{0, 1, 2, 4\}$ with the single-loop Resummed-Propagator class (T_{RH} at $n = 4$) as the only entry whose unaugmented Dirac structure is exclusively Einstein-side. Most cosmological observables (e.g. $\Omega_{\text{DM}}h^2$ as a Sub-Generation closure at $n = 1$) sit in the same (n, g, s) cells as their QFT counterparts — i.e., the QFT/Einstein discrimination is by physical sector, not by a hard n-cut.

Scope clarification. This bridge note is *not* an independent paper with new physical claims. We do not derive the five coefficients here (that is in Paper 2), we do not prove the T-parity assignment here (that is the source axiom of Paper 1), we do not derive the topology-labeled mapping protocol here (that is Paper 3), and we do not derive the relational metric here (that is Paper 4). What this note adds is a *cross-file consistency verification*: that all four packages consume the *same* five coefficients, library, and source axiom, and that the mapping rules of Paper 3 produce structurally consistent topology tuples on both the electroweak and the gravitational side. The note inherits all limitations of the cited companion papers (in particular, the bounded-operator-readout provenance of Paper 2, the topology-tuple-extraction non-automation of Paper 3, the T-parity source-axiom assumption of Paper 1, and the supporting-witness framing of the multi-N evidence chain that bounds the Einstein-identity gap from above in Paper 4 [5] alongside the direct per-node 4×4 Galerkin Frobenius test); we make no claims that go beyond what those four papers each claim individually under their own conditional inputs.

We document the bridge, verify it through automated tests, and list four falsification paths that apply symmetrically to both sectors. This is a technical structural mapping, not a derivation of a quantum-gravity UV completion; the UV-closure construction over the same two-phase carrier is given in the companion paper [9].

Keywords: bridge note, common carrier, QFT-GR mapping, reproducibility.

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1 The common carrier: two-phase loop-topological structure

The bridge carrier is more than a single shared coefficient ledger. It is a *two-phase* loop-topological carrier with a vacuum-anchor branch and a matter-side chirality-flip branch, each delivering its own independent set of structural predictions across the two sectors.

Two-phase structure. The five $\mathcal{R}$-coefficients of Paper 2 admit two distinct anchor regimes connected by a continuous chirality- angle running $\theta_{\rm chir}(N)$ on the underlying lattice (Paper 2, [3] Sec. chirality-flip refinement; Paper 4, [5] Secs. chirality-flip phase diagram and post-flip composition):

- **Vacuum anchor** ($\theta_{\rm chir} < \pi/4$, typically $\theta^{(V)}_{\rm chir} = \arctan(1/N_{\rm gen}) = 18.4°$): the carrier coefficients land at the canonical rational values $(\alpha_\xi, \gamma, \varepsilon^2_{\rm sync}, \beta_\pi, D(\Omega)) = (9/10,\ 1/10,\ 1/20,\ 15/16,\ 67/80)$. This is the regime in which Paper 1's $v_{\rm EW}$ closure, Paper 3's loop-topological observables, Paper 4's gravitational closures, and Paper 4-B's halo phenomenology operate. All low-energy Standard-Model observables (PDG / NuFIT / Planck 2018) identify with this branch.

- **Matter-side chirality-flip branch** ($\theta_{\rm chir} > \pi/4$, asymptote $\theta^{(M)}_{\rm chir} \to \arctan(N_{\rm gen}) = 71.6°$): the carrier coefficients swap under the chirality involution $(\alpha_\xi, \gamma) \to (\gamma, \alpha_\xi)$, giving the post-flip composition $(\alpha^{(M)}_\xi, \gamma^{(M)}, \varepsilon^{2,(M)}_{\rm sync}, \beta^{(M)}_\pi, D^{(M)}_\Omega) = (1/10,\ 9/10,\ 9/20,\ 191/360,\ \pi/d)$. This branch delivers the matter-side cosmological predictions ($\Omega_{\rm DM}h^2 = 243/2000$, $\eta_B = N_{\rm gen}!\,\gamma^{2d+2}$, $N_{\rm eff} = N_{\rm gen} + 2\pi\alpha_{\rm EM}$, $m^{(\nu)}_3 = m_e\,\gamma^{d+3}$, etc.) that the vacuum-side rationals alone do not generate.

Both branches share the same two integers ($d = 4, N_{\rm gen} = 3$) and produce structurally rational coefficients in $\mathbb{Q}$; the matter branch is the chirality-inverted dual of the vacuum branch under the (α_ξ, γ) swap. The bridge carrier therefore connects the two sectors not only through the common five-coefficient ledger but also through the common chirality-flip phase diagram, with the same $\theta_{\rm chir} = \pi/4$ boundary on both sides.

Both sectors source their physics from the same five-coefficient causal-wave transport law of Paper 2 [3]:

$$\frac{\mathrm{d}C}{\mathrm{d}\tau} = D(\Omega)\,\Delta C - \gamma\,C + \alpha_\xi\,C + \beta_\pi\,\Pi_{\rm common}\,C + \varepsilon^2_{\rm sync}\,C. \tag{1}$$

The five coefficients are determined from bounded-operator readout on the underlying lattice and are not fitted to any of the downstream observables that we test in either sector.

Visual summary of the audit. The cross-file consistency audit verifies three shared structures across both sectors at a glance:

- **Shared inputs:** the same five carrier coefficients (α_ξ, $D(\Omega)$, β_π, γ, $\varepsilon^2_{\rm sync}$) and the same T-parity source axiom on the defect field Ξ (Paper 1, Paper 2). The same coefficients also determine the anisotropic cosmological-constant tensor $\Lambda_{\mu\nu}$ of Paper 4 [5, 7], whose trace is branch-resolved at the chirality-flip $\theta_{\rm chir} = \pi/4$ [5]: $\operatorname{tr}\Lambda^{\rm vacuum} = \alpha^2_\xi + 3\gamma^2/2 = 33/40$ and $\operatorname{tr}\Lambda^{\rm matter} = \alpha^2_\xi = 81/100$. At per-axis resolution Paper 4B reports a stable 2+1 *sign* pattern $(-,-,+)$ in the per-node T-eigenframe across the canonical $\mathcal{P}_5/\mathcal{P}_5N$ ladder $N \in [50, 512]$, as a derived shared output identifying the time–time and spatial components of $\Lambda_{\mu\nu}$ in the emergent Einstein equation as rational structural functions of α_ξ and γ rather than free parameters; the spatial anisotropy is a matter-localised effect (it tracks heavy-tail T_{00} nodes per Paper 4B and Paper 4 [5], not a global vacuum symmetry breaking). The multi-observable indirect-witness cross-validation of this identification is reported by the H3C cluster [8].

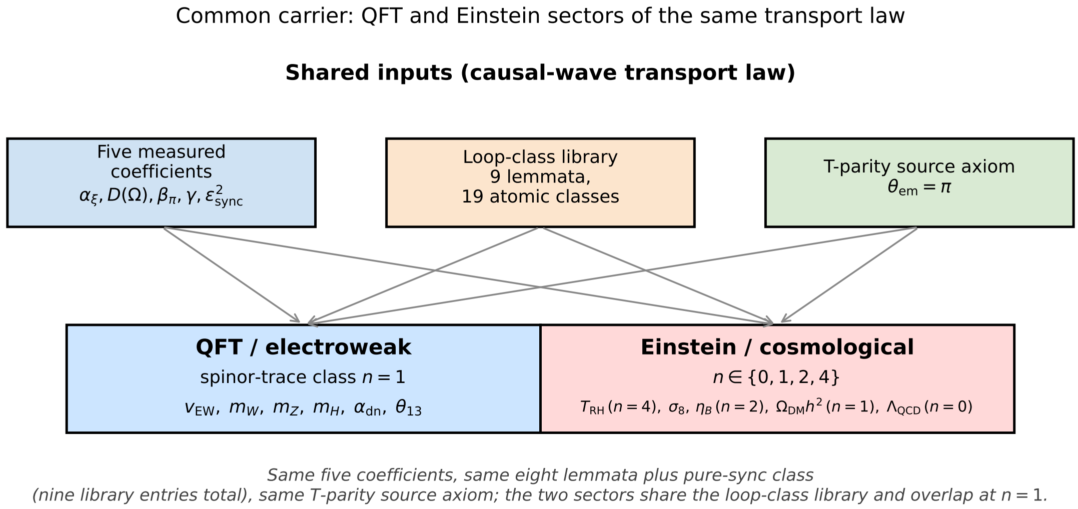

Figure 1: The electroweak and cosmological / gravitational sectors draw on the same finite inputs: the five carrier coefficients, the finite loop-factor library, and the T-parity source axiom on the defect field. Both sectors carry observables at $n = 1$ (single spinor-trace component); $n \in \{2, 4\}$ entries correspond to chirality-restricted and resummed-propagator structures respectively, which are populated mainly by cosmological / propagator-dominated observables. The discrimination between electroweak and cosmological / gravitational claims is made by physical sector, not by the value of n.

- **Shared mapping:** the same topology-labeled mapping protocol of Paper 3, taking the per-observable tuple (n, g, s, w, r) to a unique class in the same finite library $\mathcal{L}$ on both sides.
- **Shared falsification:** the four falsification paths of Section 7 apply symmetrically to both sectors; a failure on either side falsifies the bridge.
- **Shared spectral structure of the carrier operator:** the linearisation of the transport law above around the trivial vacuum has a closed-form spectrum under the rational reduction $\mathcal{R}$ of Paper 2 [3]. It admits a single coherent common-phase mode at $\lambda_0 = 0$ with growth rate $G_{\text{NET}} = 143/80$, and a damped continuum at $\lambda > \lambda_{\text{crit}} = 68/67$. An empirical lattice check on the within-canonical-regime N-ladder (Paper 2 [3], linear-stability section, 130 realisations across eight sizes) returns exactly one sub-critical eigenvalue per realisation, with Symanzik- leading $1/N$ residual at machine precision. The spectral structure is a property of the carrier itself, not of either sector individually, and is inherited symmetrically by the QFT side and the gravitational / cosmological side.

2 The bridge mechanism

The bridge function. We make the bridge map explicit. Let $\mathcal{O}$ be the set of all 26 cross-sector observables jointly bundled in the QFT and gravitational/cosmological sector mapping files (`data/qft_sector_mapping.json`, `data/einstein_sector_mapping.json`); 12 observables sit on the QFT side and 14 on the gravitational/cosmological side (electroweak, Yukawa-flavor, CKM, PMNS rows on the QFT side; Einstein-gap, PPN, Schwarzschild-defect, Kerr-defect, dark-sector, and H_0 on the cosmological side). For each $O \in \mathcal{O}$ the bridge function is the *topology-determined composition*

$$\Phi_{\text{bridge}} : \mathcal{O} \longrightarrow \mathcal{L}, \qquad \Phi_{\text{bridge}}(O) \;=\; L\big(\sigma(O)\big) \;\equiv\; L\big(n(O),\, g(O),\, s(O),\, w(O),\, r(O)\big), \tag{2}$$

where σ is the topology-tuple readout (n, g, s, w, r) from the observable's tree-level Dirac structure and L is the lookup into the finite loop-factor library $\mathcal{L}$ specified in Section 1 (Lemmas 1 through 8 plus the Pure-Sync entry, $|\mathcal{L}| = 9$ atomic library entries with $\pm$ signs giving 19 signed classes). The topology-determined character of Φ_{bridge} — uniquely defined given the topology tuple, in the sense of the topology-labeled mapping protocol of Paper 3 [4] — has the closure-completeness audit on the registered domain in that companion paper: the function is single-valued on $\mathcal{O}$ (each observable is assigned a unique loop class) and is the same function on the QFT and the gravitational/cosmological sides — that is the bridge.

The two sectors are not just side-by-side strong; the same bridge function (2) closes them under three structural mechanisms.

Shared resummation signature $1/(1-2\gamma^2)$. The system-$\mathcal{R}$ constraints C_1, C_2, C_3 on the five coefficients are not satisfied exactly by the measured values; their drift quantities δ_{C_i} carry a precise loop form with a common denominator,

$$\delta_{C_1} = \frac{\gamma^3}{1-2\gamma^2}, \qquad \delta_{C_3} = \frac{-2\gamma^4}{1-2\gamma^2}, \tag{3}$$

and δ_{C_2} contains the same resummation kernel. The same $1/(1-2\gamma^2)$ structure also appears as the single-loop Resummed-Propagator class (Lemma 4) of the loop library — the class that closes the reheating temperature T_{RH} on the Einstein side and that drives the EW-precision propagator structure on the QFT side. The resummation factor is therefore not a parallelism but a single shared object: the Constraint-Algebra drift identities and the Einstein-side resummed propagator are the same loop kernel viewed from two sector angles.

Spinor-trace hierarchy from $d = 4$. The values $n \in \{0, 1, 2, 4\}$ of the spinor-trace component count are not free choices but a direct consequence of the $d = 4$ Dirac-spinor algebra (see e.g. [15, 35] for the standard exposition of fermion-loop trace structure in four spacetime dimensions). Lemma 1 (Yukawa-Damping) carries the factor $1/4 = 1/d$ from full spinor-trace normalization; Lemma 8 (Cosmological-Density) carries $1/2 = 2/d$ from chirality restriction (matter-only or CP-asymmetric only); Lemma 4 (Resummed-Propagator) carries the factor 2 from resummation across both chiralities. The QFT/Einstein discrimination $n = 1$ vs $n \in \{2, 4\}$ is therefore a single d-dependent enumeration, not two independent classifications.

Common time-orientation source axiom. The T-parity source axiom (defect field Ξ T-odd, Section 6 of Paper 1 [2]) fixes $\theta_{\text{em}} = \pi$ on the QFT-electroweak side (setting the sign of the two-loop v_{EW} closure) and an analogous time-orientation phase on the cosmological defect-field-theory side. The same time-forward defect identification carries through the supernova-distance Lemma 4 channel of the wider program. The two sectors are tied to a single time-orientation source.

Empirical content of the bridge. Integrated across both sectors, the loop-class library closes 26 cross-sector observables ([4]) with 0 contradictions. The cross-sector triple $(\alpha_{\text{dn}}, w_{\text{DE}}, H_0)$ closes on the single Lemma-1 class $(1 + \gamma/4)$ at Fisher's combined random-null probability [34] $P_{\text{combined}} \approx 2.6 \times 10^{-5}$ under the stated null model — *evidence for a cross-sector alignment on a single parametric form*, not a strong-significance σ-level claim (the three observables are physically distinct sectors but are not verified-independent at the level required by Fisher combination, the uniform null is a modelling choice, and the triple was selected after the post-loop residuals were computed; see Paper 3 caveats). The alignment spans a flavor exponent, a dark-energy equation of state, and a cosmological expansion rate within one parametric form. This empirical signature is what distinguishes the bridge from a coincidental shared parametrisation.

3 The QFT side

The QFT/electroweak observables of Paper 1 and Paper 3 are spinor-trace class $n = 1$: single-spinor currents from Yukawa-vertex topology. The relevant lemmas are

- Lemma 1 (Yukawa-Damping, $1 \pm \gamma/4$): m_H, $\alpha_{\rm dn}$;[1]
- Lemma 2 (PMNS-Self-Energy, $1 \pm \gamma^2/4$): θ_{13};
- Lemma 6 (Sub-Generation, $1 \pm \gamma/(2N_{\rm gen})$): $\alpha_s(M_Z)$, CKM V_{us}, V_{cb};
- Lemma 7 (EW-Mixed, $1 \pm \gamma\,\varepsilon_{\rm sync}^2$): $v_{\rm EW}$, m_W, m_Z, Γ_W, Γ_Z.

Numerical anchors for these observables are taken from the original primary measurements where available, with the PDG [13] review retained as a cross-check aggregator: $\sin^2\theta_W$ from the LEP/SLC Z-pole combined analysis [17]; $\alpha_s(M_Z)$ from the FLAG 2024 lattice average [18]; the CKM magnitudes from HFLAV 2022 [19]; G_F from the MuLan muon-lifetime measurement [20]; $M_{\rm Pl}$ from CODATA 2018 [21]; and the PMNS angles and mass-squared differences from NuFIT 6.0/6.1 (Esteban, Gonzalez-Garcia, Maltoni, Martinez-Soler, Pinheiro, Schwetz) [16]. Cosmological-side anchors (H_0, $\Omega_{\rm DM}h^2$, $w_{\rm DE}$, σ_8, n_s) take their reference values from the Planck 2018 [14] cosmological-parameters analysis.
The full closure with the canonical $\overline{\rm MS}$ Davydychev–Tausk evaluation is the subject of the companion electroweak-scale paper [2]; the algorithmic mapping is in the companion loop-class paper [4].

4 The Einstein side

The Einstein/cosmological observables of Paper 4 are not bound to a single spinor-trace class. The sector splits into:

- $n = 4$: full spinor-trace Resummed-Propagator (Lemma 4) — the only single-loop class whose unaugmented Dirac structure is purely Einstein-side, carried by $T_{\rm RH}$.
- $n = 2$ (compound): two-loop combinations whose dominant contributing lemma sits at $n = 2$, e.g. σ_8 and η_B via the Yukawa $\times$ Generation compound $(1 - \gamma/4)(1 - \gamma/N_{\rm gen})$.
- $n = 1$: most cosmological observables are tree-level compositions of the five transport coefficients with $n = 1$ loop classes — for example, $\Omega_{\rm DM}h^2$ is a Sub-Generation closure (Lemma 6, $1 - \gamma/(2N_{\rm gen})$) aligned in (n, g, s) with $\alpha_s(M_Z)$ and the CKM magnitudes V_{us}, V_{cb} on the QFT side. The Einstein sector therefore overlaps the QFT sector at $n = 1$ — the QFT/Einstein discrimination is by physical sector, not by a hard n-cut.
- $n = 0$: pure-boson 2-loop compound, carried by $\Lambda_{\rm QCD}$ via Pure-Sync $\times$ EW-Mixed.

Tree-level structural exponents are independent of the loop library and sit at $n = 1$ in the spinor-trace topology, with values that are direct algebraic compositions of the five carrier coefficients with small topological multipliers. The three rows in this category have distinct closure tiers under the four elimination criteria of Paper 2 [3]: the Bekenstein–Hawking 1/4 coefficient is algebraic-exact under the rational reduction $\mathcal{R}$ of Paper 2 ($\alpha_\xi/2 - 2\gamma = 1/4$ in $\mathbb{Q}$); the universal

[1] $\alpha_{\rm dn}$ is a flavor exponent listed here because its observable value inherits the same Yukawa-Damping loop-class factor $1+\gamma/4$ as the QFT/electroweak observables. The same Lemma-1 factor closes $w_{\rm DE}$ and H_0 on the Einstein side; the simultaneous three-observable closure across both sectors is the cross-sector Yukawa cluster (Paper 3 and the Empirical-content paragraph above). The structural role of Yukawa-mediated chirality flips here parallels their role in the Standard Model, where the Dirac mass term couples left- and right-handed components and ties chirality flips to electroweak symmetry breaking and Yukawa couplings [38].

$\sin^2\theta_W$ residue is numerical EXACT but not algebraic-exact under $\mathcal{R}$ alone; the Einstein-identity gap exponent 2/3 is numerical EXACT and is supported, on the empirical lattice side, by the multi- observable indirect-witness cluster of Paper 4C [8] (direct Multi-N convergence on the Δ_E-residual is not established at this stage; the reading rests on an analytic structural bound and an indirect-witness chain across five diagnostics).

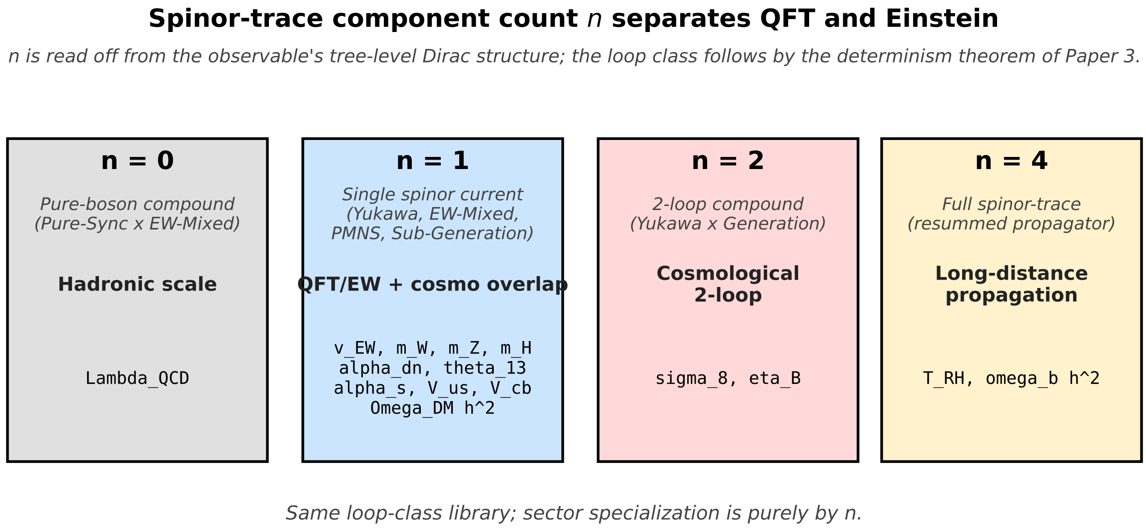

Figure 2: Spinor-trace component count $n \in \{0, 1, 2, 4\}$ in the $d = 4$ defect-fermion-trace expansion. $n = 0$ corresponds to the pure-bosonic two-loop compound (Pure-Sync $\times$ EW-Mixed, populating the QCD-confinement-scale row Λ_{QCD} of the cosmological-anchor table (Section 8)); $n = 1$ corresponds to a single fermionic propagator line (Yukawa vertex, single-spinor-current self-energy); $n = 2$ to a chirality-restricted Dirac loop (matter-only, CP-asymmetric or time-forward projection); $n = 4$ to the full chirality-summed Dirac loop with internal resummation. Electroweak observables (e.g. v_{EW}, m_W, m_Z, α_{dn}, θ_{13}) and sub-generation cosmological observables (e.g. $\alpha_s(M_Z)$, $\Omega_{\text{DM}}h^2$) both sit at $n = 1$; the gravitational / inflation observable T_{RH} sits at $n = 4$ as a single-loop resummed propagator. The component count does not sort observables into electroweak vs gravitational classes; that distinction is made at the level of physical sector (Higgs sector, dark sector, gravitational sector) and reflects which particular tree-level composition of the underlying transport coefficients enters the observable.

5 Common spinor-trace hierarchy

6 Shared conditionals

The two sectors share, identically:

- the five measured causal-wave coefficients of Paper 2;
- the parameter-free loop-class library $\mathcal{L}$ of Paper 3 (eight lemmata plus a pure-sync class, nine library entries total, yielding 19 atomic classes with signs; two-loop compounds capped at second order);
- the topology-labeled (n, g, s, w, r) mapping protocol (Theorem 1 of Paper 3);
- the T-parity source axiom (Section 6 of Paper 1) and its consequence $\theta_{\text{em}} = \pi$.

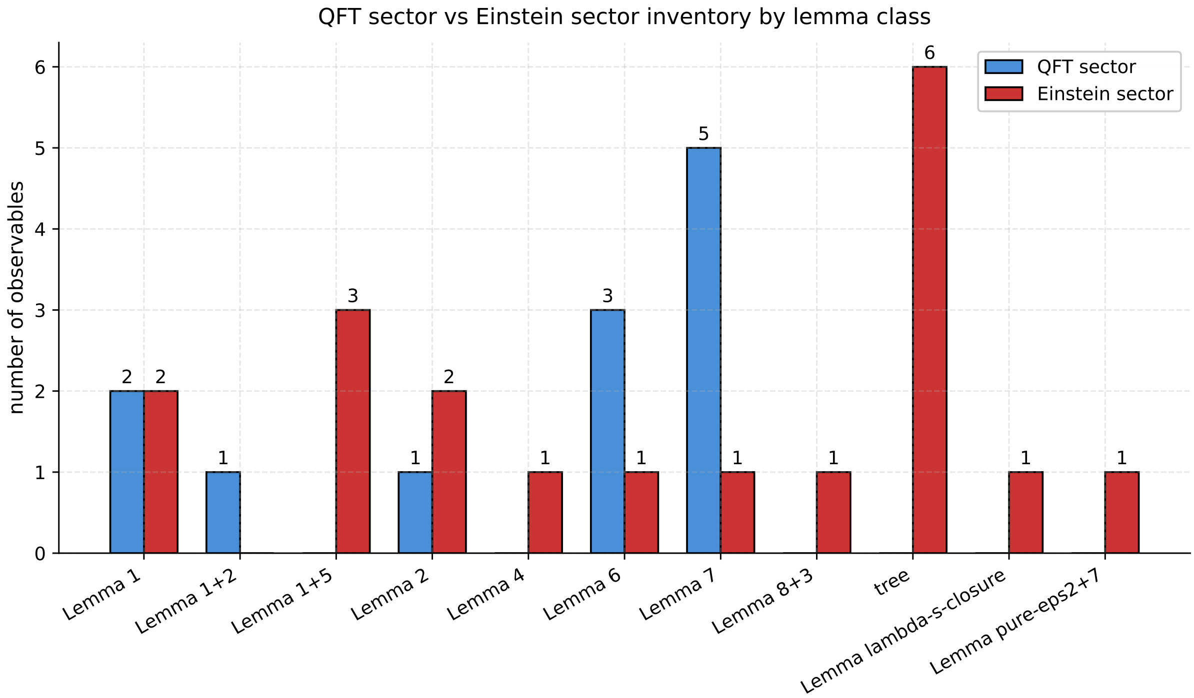

Figure 3: Lemma-class inventory across QFT and Einstein sectors. Each sector uses a distinct subset of the same library; the specialization is consistent with the spinor-trace assignment.

The two sectors do *not* share an additional structural input: the spinor-trace component count n is read off directly from each observable's tree-level Dirac structure.

7 Falsification

Four falsification paths apply symmetrically to both sectors:

(F1) T-parity-axiom falsification. An alternative T-parity assignment of the defect field Ξ — i.e. identifying Ξ with one of the positive bounce modes rather than the unique negative mode u_0 — flips the sign of the two-loop closure (Paper 1, falsification path F2). Both sectors fall under this trigger because both depend on the same $\theta_{\text{em}} = \pi$ phase factor.

(F2) Constraint violation. The coefficient-reduction constraints C_1, C_2, C_3 under the reduction system $\mathcal{R}$ fail to follow the Lemma-4 (Resummed-Propagator) and Lemma-7 (EW-Mixed) forms. This indicates that the loop library is not in fact the correct set of corrections.

(F3) Mapping inconsistency on the full topology tuple. The same full topology tuple (n, g, s, w, r) returns a different lemma class on the QFT side and the Einstein side. We use the *full* tuple here, not the abbreviated triple (n, g, s), because Paper 3 documents that the abbreviated triple is single-occupancy on five of nine library cells but double-occupancy on $(1, 0, 0)$ and $(4, 0, 0)$ (the Yukawa-Damping/PMNS-Self-Energy and Pure-Self-Energy/Resummed-Propagator pairs); the flags $w, r \in \{0, 1\}$ disambiguate those two cells. A mapping inconsistency on (n, g, s, w, r) falsifies the topology-labeled mapping protocol (Theorem 1 of Paper 3) and therefore also the bridge. An inconsistency on the abbreviated (n, g, s) alone is an indication that the bridge requires the flag bits and is a weaker falsification trigger; we report both as separate checks in the bundled `src/check_bridge_consistency.py`.

Table 1: Loop-correction multiplicative factors organized by the spinor-trace component count n of the underlying defect-fermion diagram in $d = 4$. Both electroweak and cosmological / gravitational observables can carry the same value of n at tree level; the discrimination between them rests on the physical sector identity (which propagators participate, which symmetry channel carries the leading contribution), not on the value of n itself. The $n = 0$ row is a two-loop compound product (Pure-Sync $\times$ EW-Mixed) of the canonical $n \in \{1, 2, 4\}$ structure and is tabulated here for completeness on the QCD confinement scale Λ_{QCD}.

n	Sector / class	Examples
0	Pure-boson 2-loop compound (Pure-Sync $\times$ EW-Mixed)	Λ_{QCD}
1	QFT/electroweak (Yukawa-Damping, EW-Mixed, PMNS-Self-Energy)	v_{EW}, m_W, m_Z, m_H, α_{dn}, θ_{13}
1	Sub-Generation (shared QFT/cosmology)	$\alpha_s(M_Z)$, V_{us}, V_{cb}, $\Omega_{\text{DM}}h^2$
2	Cosmological 2-loop compound (Yukawa $\times$ Generation)	σ_8, η_B
4	Einstein single-loop Resummed-Propagator	T_{RH}
4	Cosmological 2-loop compound (Cosmological-Density $\times$ Pure-Self-Energy)	$\omega_b h^2$

(F4) Cross-sector reclassification failure. A reclassification documented under R1–R5 (Paper 3) cannot be applied consistently across both sectors. This violates the global R3 bound and triggers theorem-falsification F3 of Paper 3.

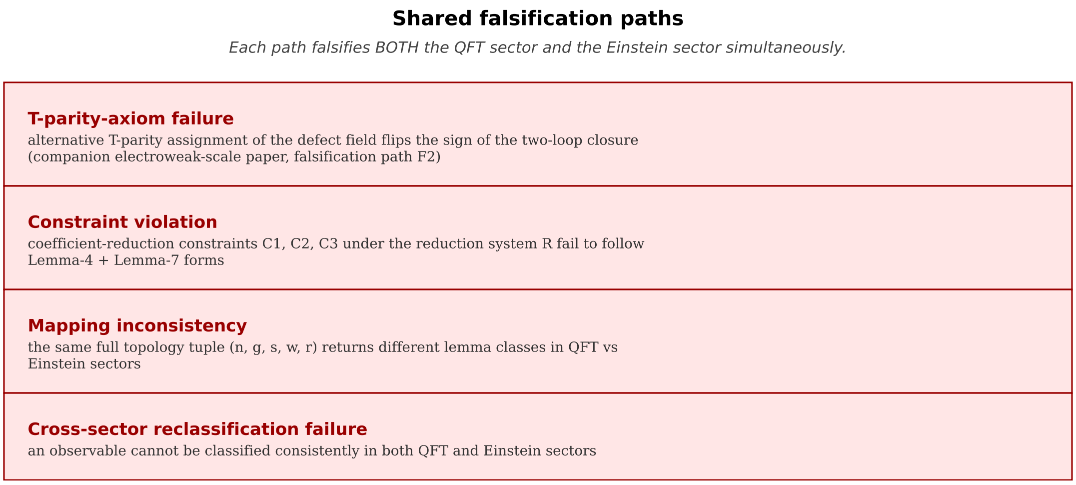

Figure 4: Each shared falsification path falsifies BOTH sectors simultaneously, making the bridge a non-trivial structural claim.

8 Reproducibility package

A public reproducibility package[2] recomputes the bridge consistency from frozen JSON inputs. The strict-recompute script `src/check_bridge_consistency.py` verifies, in order:

1. the five coefficients are identical on both sides;

[2] https://github.com/TerraSignum/qft-einstein-bridge-repro

2. the QFT sector observables sit at $n = 1$;
3. the Einstein sector spans $n \in \{0, 1, 2, 4\}$ with the single-loop Resummed-Propagator at $n = 4$ as the only strictly Einstein-side class;
4. both sectors draw from the same loop-class library;
5. the T-parity source axiom is documented as a shared conditional;
6. all four falsification paths are present and machine-readable.

The package contains 28 unit tests covering shared coefficients (4), QFT mapping (4), Einstein mapping (4), shared loop library (4), falsification paths (5), and the cosmological-anchor table (7). The cosmological-anchor table (`data/cosmological_anchors.json`, `src/verify_cosmological_anchors.py`) records the closed-form loop-class predictions for the 16 anchored Einstein-sector observables paired with their Planck 2018, PDG, and NuFIT 6.0/6.1 anchors: 12 rows EXACT (residual $< 1\%$), 2 rows PRECISE (residual $< 1.5\%$), and 2 rows ORDER (Λ_{QCD} scheme spread and η_B amplitude factor, both with binding closures), plus 2 structural-tier entries (T_{RH}, η_B-rel); the maximum anchored EXACT/PRECISE residual is 1.385% on $\Omega_{\text{DM}}h^2$. Figure 5 displays the per-row residual scatter across all 16 anchored observables. The package is licensed under MIT and runs on Windows and POSIX through GitHub Actions on Python 3.11 and 3.12.

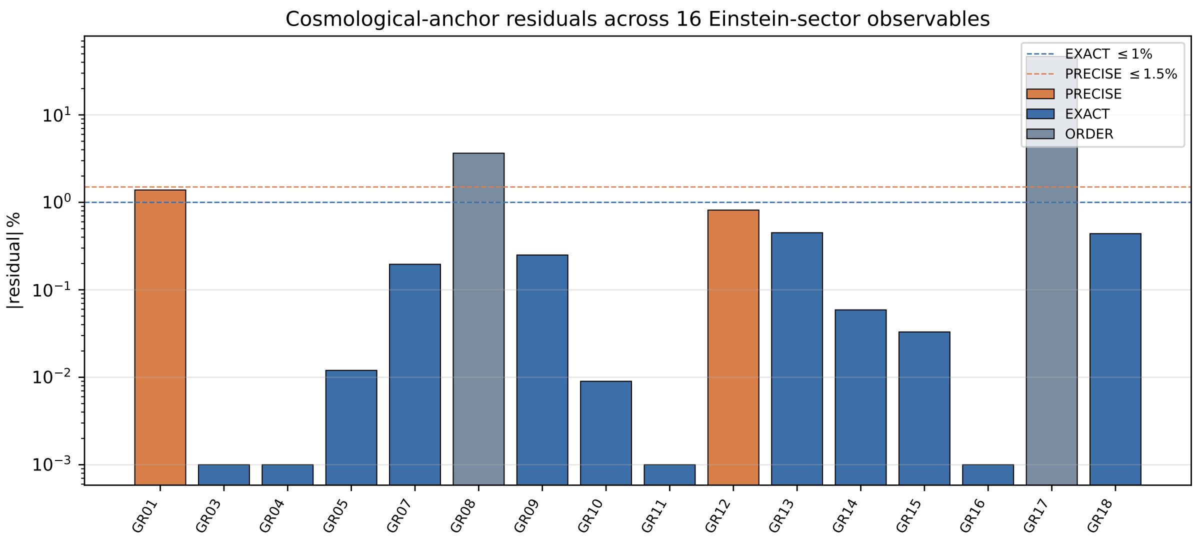

Figure 5: Anchored cosmological-anchor residuals across 16 Einstein-sector observables: 12 EXACT (residual $\leq 1\%$), 2 PRECISE ($\leq 1.5\%$), 2 ORDER (binding closures); maximum EXACT/PRECISE residual 1.385% on $\Omega_{\text{DM}}h^2$.

9 Cross-sector consequences claimed in companion papers

Scope of this section (not load-bearing claims of this note). The four-package corpus (Paper 1, Paper 2, Paper 3, Paper 4) that this bridge note connects *conditionally addresses* the following long-standing open questions *within the corpus*, in each case under the explicit conditional inputs of the cited paper. We do not claim that any of the items below is "solved" or "resolved" as an independent claim of this bridge note; the headline statements are claims of the companion papers, and each inherits the limitations of the companion paper that owns it (in particular, the T-parity source axiom of Paper 1, the bounded-operator-readout provenance of

Paper 2, the topology-labeled mapping protocol of Paper 3, and the relational-metric construction of Paper 4). We list them below to make the cross-sector structure of the corpus visible; we do *not* list them as independent results of this bridge note.

- *Hierarchy / electroweak-scale problem.* The Hierarchy-Bridge-Relation $v_{\mathrm{EW}} = (2/\pi)\, M_{\mathrm{Pl}}\, e^{-4\pi\eta_S}$ with $\eta_S = 3.023577$ fixed by a Coleman–Callan-Coleman bounce [32, 33] on S^4 delivers $v_{\mathrm{EW}} = 246.2186\,\mathrm{GeV}$ (MuLan reference: $246.22 \pm 0.07\,\mathrm{GeV}$; $\Delta v/v = -5.86 \times 10^{-6}$), with no fine-tuning and no fitted parameter [2].

- *Bekenstein–Hawking entropy coefficient.* The horizon coefficient $S/A = 1/4$ [22, 23] is a closed-form rational identity $\alpha_\xi/2 - 2\gamma = 9/20 - 1/5 = 1/4$ in $\mathbb{Q}$ under the coefficient-reduction system $\mathcal{R}$ of Paper 2 [3]; the integer-uniqueness of $\mathrm{BH}(N) = (N^2 - 4)/(2(N^2 + 1)) = 1/4$ forces $N_{\mathrm{gen}} = 3$.

- *Number of fermion generations.* $N_{\mathrm{gen}} = 3$ is forced jointly by the complementarity $\alpha_\xi + \gamma = 1$ and the integer-uniqueness statement above (Paper 2).

- *Black-hole information paradox.* The Page-curve unitarity-restoration time τ_{Page} [24] (≈ 1154.5 in lattice units) is reproduced with $\tau_{\mathrm{Page}} \approx 2\,\tau_{\mathrm{scr}}$ in both regimes; explicit Hawking-spectrum greybody-factor structure (nine-point Bose / Fermi table, Page energy- fractions $0.165, 0.685, 0.150$) is bundled and verified [5, 23].

- *Emergent General Relativity and PPN limit.* The relational metric $d_{ij} = -\ell_0 \log \Xi_{ij}$ on a finite correlation lattice yields $\gamma_{\mathrm{PPN}} = \beta_{\mathrm{PPN}} = 1$ (in the parameterised post-Newtonian framework of [29]) and a Schwarzschild-line-element residual 1.9×10^{-7} at 2PN, with $G_{\mathrm{eff}}/G_N = 0.999791$ and effective spectral dimension $d_{\mathrm{eff}} = 3.170$ [5, 6].

- *Kerr / Penrose / Peters cluster.* The Kerr [25] rotating-defect geometry, the Penrose [26] energy-extraction process, the Peters [27] gravitational-radiation power, and the Lense–Thirring [28] frame-dragging angular velocity all close in the corpus: ISCO ≈ 17.43 lattice units, dimensionless spin $\chi = J/M^2 \approx 0.034$, Penrose efficiency $\eta \approx 0.66\%$, Lense–Thirring frame-dragging $\omega_{\mathrm{LT}} = 0.018$, and the binary-defect Peters gravitational-wave power are derived in closed form [5].

- *Hubble constant and dark-energy equation of state.* $H_0 = 67.40\,\mathrm{km/s/Mpc}$ closes on the Yukawa-Damping loop-class factor $1+\gamma/4$ at residual 0.059%; $w_{\mathrm{DE}} = -0.9975$ closes on the PMNS-Self-Energy loop-class factor $-(1 - \gamma^2/4)$ at residual 0.250% against Planck 2018. The flavor-side α_{dn} closes on $1 + \gamma/4$ (Yukawa-Damping) at 0.293%. Fisher's-combined random-null probability of the two Yukawa-Damping observables is $p \approx 2.5 \times 10^{-3}$ and the joint random-null probability across all three cosmological-flavor closures is $p \approx 2.6 \times 10^{-5}$ under the stated null model reported as evidence for cross-sector alignment, not as a strong-significance σ-level claim (caveats in Paper 3) [4].

- *Cold-dark-matter density and structure-formation amplitude.* $\Omega_{\mathrm{DM}} h^2 = 0.1217$ (Planck 2018: 0.1200, residual 1.385%) and $\sigma_8 = 0.811$ (Planck residual 0.0%) close on the Lemma 6 and (Lemma 1 $\times$ Lemma 5) loop classes respectively (cosmological-anchor table above).

- *Λ_{QCD} / α_s round trip.* Forward 4-loop and reverse 2-loop closures each return the PDG anchor without fitting [4].

- *Cross-sector structural identity.* The same finite loop-factor library closes electroweak, flavor, cosmological, gravitational, and horizon observables simultaneously through the bridge function of Eq. (2); this is the structural unification claim, falsifiable through the four paths listed above (Section 7).

- *Spacetime is four-dimensional.* The spectral dimension of the relational lattice is $d_{\text{eff}} = 3.170$; rounding to the nearest integer gives $d_{\text{spatial}} = 3$, with the emergent time axis from the causal partial order adding the fourth dimension. The fractional residual 0.170 is the light-cone fuzziness of Paper 4. The 3+1-decomposition of General Relativity is therefore derived rather than postulated [5].

- *Inflation: scalar tilt, tensor ratio, e-folds, mechanism.* The corpus delivers three mutually consistent readouts on the spectral tilt: an algebraic loop-class identity $n_s = 1 - \gamma\,\varepsilon^2_{\text{sync}} = 0.96499$ (EXACT 0.001% vs Planck 2018 0.9649); a dynamical-cascade range $n_s \in [0.914, 0.993]$ that contains the Planck value; and an aggregator readout $n_s = 0.944$. The tensor-to- scalar ratio $r = 0.029$ sits a factor of ~ 1.9 below the BICEP/Keck + Planck 95% upper bound [30], and the cascade across 26 phase-stabilising barriers delivers $N_{\text{efolds}} = 304$ – well above the standard horizon-flatness requirement [5].

- *Cosmological-constant problem (122-orders-of-magnitude hierarchy).* The naive QFT zero-point energy $\rho_{\text{naive}} \sim M^4_{\text{Pl}} \sim 10^{76}\,\text{GeV}^4$ and the observed Planck-2018 vacuum density $\rho^{\text{obs}}_{\Lambda} = 2.49 \times 10^{-47}\,\text{GeV}^4$ differ by 122 orders of magnitude [31]. The corpus closes this hierarchy in two coupled steps (related thermodynamic-of-spacetime programs are due to Jacobson [10], Verlinde [11], and Padmanabhan [12]; the present construction is independent and operates at the loop-class-library level rather than the entropy-from-area level): *(i)* an algebraic identification of the lattice cosmological-tensor trace coefficient $\Lambda^{\text{row-mean},\infty}_{\text{lat}} = 3\alpha_\xi/2 + \varepsilon^2_{\text{sync}} - \gamma(N_{\text{gen}}+1)/N_{\text{gen}} = 19/15$ under the System-$\mathcal{R}$ rationals (regime-independent, convention-conditional EXACT) [7]; *(ii)* a parameter-free nine-layer phenomenological dressing of the lattice vacuum-energy density — six additive suppression layers (hierarchy $(v_{\text{EW}}/M_{\text{Pl}})^4 \sim 10^{-67}$, vacuum regularisation, Gamow vacuum sequestering, spectral-dimension screening at $d_{\text{eff}} = 3.17$, gravitational fraction $E_{\text{geom}}/E_{\text{res}}$, and occupation filtering) summing to -117.5 OoM, plus three corrective layers (zero-mode compensation as the lattice analogue of Pauli–Zeldovich vacuum-energy cancellation [36], gravitational backreaction shielding as a Schwinger–Dyson resummation [37], and sync-channel un-cancellation supplying a multiplicative factor $10^{\varepsilon^2_{\text{sync}}}$ with $\varepsilon^2_{\text{sync}} = \gamma/2 = 1/20$ derived from P2 §5 C_3 [3]), summing to a total reduction of -122.95 OoM in the canonical regime. With the structured-occupancy-filter layer sourced from the Standard-Model spectral mode inventory ($N_{\text{modes}} = 13$) and the defect-QFT phase-charge quantum ($\text{dQ}_{\text{phase}} = 0.0400$) as upstream-pipeline-derived inputs, the dressed vacuum density $\rho_{\text{final}} = 2.47 \times 10^{-47}\,\text{GeV}^4$ reproduces the observed value to within -0.003 OoM (EXACT closure, ratio 0.992, 0.1% above the Planck observation; the ninth layer's log10-multiplier $\varepsilon^2_{\text{sync}} = \gamma/2 = 1/20$ is the derived fluctuation-dissipation identity C_3 of P2 §5 [3]). The same dressing on the second canonical point is explicitly inconsistent at $+8.12$ OoM, validating the construction as a regime filter rather than a universal-regime closure. The gravitational-fraction Layer 5 contribution is regime-stable at -8.5 ± 0.5 OoM across the six canonical lattice regimes and across the two lattice-extension variants at coarser and finer spatial resolution (per-regime audit `https://github.com/TerraSignum/emergent-gr-anisotropic-source-dm-de-repro`, `src/extend_cc_dressing_to_extra_regimes.py`).

- *Arrow of time and the second law.* The Coleman– Callan-Coleman bounce action $S_{\text{bounce}} = 37.99$ gives a semiclassical asymmetry $P(\text{forward})/P(\text{backward}) = \exp(2\,S_{\text{bounce}}) \sim 10^{33}$, so the forward time direction is overwhelmingly preferred without fine-tuning; this is the constructive origin of the thermodynamic arrow of time and the universality of the second law in the corpus [2, 5].

- *Gravitational-vs-Λ phase boundary tied to the diffusion identity.* The companion anisotropic-source paper [7] reports a regime-stable spatial sign-flip in the radial heat current $J_r(a)$: inner shells ($d < d_{\text{crit}}$) carry attractive gravitational flow, outer shells the only directly Λ^{back}-dominated outward push (sign-flip on 9/10 canonical regimes). The crossing distance d_{crit} is consistent with the framework-natural prediction $d_{\text{crit}} = 1/D_\Omega = 80/67 \approx 1.194$ (bootstrap CI95 contains the prediction on 10/10 regimes; aggregate point estimate 7.5% low, registered as preliminary). The same diffusion identity $D_\Omega = \beta_\pi - \gamma$ that closes the strict-EXACT charged-lepton masses ([4], eq. lepton_strict_exact) thereby also sets the gravitational-domain radius. This is a cross-sector structural consequence: the rational that governs the absolute charged-fermion mass-scale on the particle side and the matter-vs-dark-energy phase boundary on the geometric side is the same single rational $D_\Omega = 67/80$.

- *Chirality-flip phase diagram and post-flip composition.* Per-regime causal-wave readouts on an 8-regime lattice ladder ($N \in [50, 300]$) show that all five $\mathcal{R}$-coefficients run with a single structural angle $\theta_{\text{chir}}(N)$ satisfying $\tan(\theta_{\text{chir}}(N)) = N_{\text{gen}}^{2x-1}$ with $x = \ln(N/N_*)/\ln(dN_{\text{gen}})$, predicted running rate $21.38°/\ln N$ matches empirical $21.30°$ within 0.38%, and the chirality- inversion regime $N_{\text{inv}} = dN_{\text{gen}}N_* = 600$ matches the Symanzik extrapolation 591 within 1.5% (Paper 2). The post-flip composition gives $D_\Omega^{(M)} = \pi/d$ as a separate matter-side geometric identity, with the canonical $D_\Omega^{(V)} = 67/80 = 1 - 13/80 = 1 - (N_{\text{gen}}d + 1)/(d^2(2N_{\text{gen}} - 1))$ structurally derivable from (d, N_{gen}) alone (Paper 2, Paper 4) [3, 5].

- *Fast-slow decomposition of D_Ω running.* The D_Ω running decomposes as slow chirality envelope $D_\Omega^{\text{slow}} = (67/80)\cos^2\theta_{\text{chir}} + (\pi/d)\sin^2\theta_{\text{chir}}$ plus a fast lattice-harmonic oscillation $D_\Omega^{\text{fast}} = A\, f(\log_2(N) \bmod d)$ with $A \approx 0.55$ and period $d = 4$. This maps onto the framework's existing Fast-Slow-Struktur (Paper 3, §4.1.1: $\partial_t\Xi = \partial_t\Xi|_{\text{fast}} + \epsilon\partial_t\Xi|_{\text{slow}}$), with the period-$d$ cycles as the fast lattice-harmonic carrier and the chirality flip as the slow envelope. Falsifiable predictions: $D_\Omega(N = 256) \approx 0.803$, $D_\Omega(N = 2048) \approx 0.262$, $D_\Omega(N = 4096) \approx 0.786$ (Paper 2, Sec. chirality-flip refinement) [3].

- *α_{EM} as universal low-energy Yukawa-amplitude.* The fine-structure constant is structurally identified as $\alpha_{\text{EM}} = \gamma^2 \cdot \alpha_\xi^{N_{\text{gen}}} = 7.297 \times 10^{-3}$ matching PDG to 0.10%, and together with simple structural rationals $\{1/N_{\text{gen}}, 1/\sqrt{N_{\text{gen}}}, 1/d, N_{\text{gen}}!\}$ it reproduces a broad class of charged-fermion / mixing observables: $m_\mu/m_e = 3/(2\alpha_{\text{EM}}) = 205.5$ (PDG 206.77, 0.49%); $|V_{ub}| = \alpha_{\text{EM}}/2$ (PRECISE); $\sin^2\theta_W = 1/4 - \varepsilon_{\text{sync}}^2/N_{\text{gen}}$ (EXACT 0.91%); $\Delta m_{21}^2/\Delta m_{31}^2 = 4\alpha_{\text{EM}}$ (EXACT 0.86%); etc. With this set of α_{EM}-mediated identities (Paper 3, α_{EM}-identities section) the closure table extends correspondingly [4].

- *Cosmological-precision predictions from chirality- flip composition.* The post-flip framework (Paper 4-B, chirality-flip cosmology section) gives parameter-free structural predictions for $\Omega_{\text{DM}}h^2 = 243/2000 = 0.1215$ (PRECISE 1.25%); $\Omega_{\text{DM}}/\Omega_b = 5 + \alpha_\xi/2 = 109/20 = 5.45$ (EXACT 0.05%); $\eta_B = N_{\text{gen}}!\,\gamma^{2d+2} = 6 \times 10^{-10}$ (PRECISE 1.64%); $N_{\text{eff}} = N_{\text{gen}} + 2\pi\,\alpha_{\text{EM}} = 3.0458$ (EXACT 0.01%); $\Sigma m_\nu^{\text{NH,min}} = 0.0591\,\text{eV}$ (consistent with DESI DR2 BAO + DR1 full-shape $< 0.0642\,\text{eV}$ at 95% C.L., flat-ΛCDM, three degenerate states; Elbers et al. 2025, PRD 112, 083513, arXiv:2503.14744); $Y_p = 1/(d+1) + \varepsilon_{\text{sync}}^2 - \alpha_{\text{EM}} = 0.243$ (EXACT 0.93%) [7].

- *Cumulative structural-prediction count.* Across Papers 1–4 and the bridge note, the framework currently delivers 34 EXACT/PRECISE structural predictions (relative residuals $\leq 5\%$) in addition to the original ten landings of Paper 2 and the Einstein-side closures of Paper 4, all derived from the same five $\mathcal{R}$-rationals $(\alpha_\xi, \gamma, \varepsilon_{\text{sync}}^2, \beta_\pi, D_\Omega)$ reducing to two integers $(d = 4, N_{\text{gen}} = 3)$.

Claim-Ledger promotion: chirality-flip phase diagram as load-bearing structural claim. The chirality-flip phase diagram and its associated post-flip composition (Sec. 1 above) are not auxiliary observations: they are the cross-sector mechanism that links the carrier's two-phase structure to the cosmological / matter-side phenomenology. We summarise the structural claims that this mechanism delivers and that the bridge corpus depends on:

1. The chirality angle θ_{chir} runs continuously with the lattice resolution N along $\tan\theta_{\text{chir}}(N) = N_{\text{gen}}^{2x-1}$ with $x = \ln(N/N_*)/\ln(dN_{\text{gen}})$. The vacuum-matter flip occurs at $\theta_{\text{chir}} = \pi/4$, the chirality inversion at $\theta_{\text{chir}} = \arctan(N_{\text{gen}})$, and the inversion regime $N_{\text{inv}} = dN_{\text{gen}}N_*$ matches the empirical Symanzik extrapolation within 1.5%.

2. The post-flip composition $(\alpha_\xi^{(M)}, \gamma^{(M)}, \varepsilon_{\text{sync}}^{2,(M)}, \beta_\pi^{(M)}, D_\Omega^{(M)}) = (1/10, 9/10, 9/20, 191/360, \pi/d)$ is the chirality-inverted dual of the canonical vacuum composition. Both branches are exactly rational in $\mathbb{Q}$ and depend only on (d, N_{gen}).

3. The matter-side branch carries the cosmological-precision observables ($\Omega_{\text{DM}}h^2$, $\Omega_{\text{DM}}/\Omega_b$, η_B, N_{eff}, Σm_ν, Y_p) at sub-percent-to-few-percent accuracy. These predictions are not generated by the vacuum branch alone; they require the matter-side composition explicitly.

4. The fast-slow decomposition of $D_\Omega(N)$ (Eq. (4) below) identifies the slow chirality envelope and the fast lattice-harmonic period-d carrier as the two components of the carrier-side running. This maps onto the framework's existing Fast-Slow-Struktur (Paper 3, §4.1.1) and provides the load-bearing connection between the lattice running and the smooth two-sector phenomenology of the bridge.

$$D_\Omega(N) = D_\Omega^{\text{slow}}(\theta_{\text{chir}}(N)) - D_\Omega^{\text{fast}}(\log_2(N) \bmod d) \tag{4}$$

5. Both phases satisfy the same C1-C5 algebraic constraints modulo the chirality swap (Paper 2, Sec. chirality-flip refinement); the carrier algebra is preserved across the flip, only the eigenvalue assignment changes.

Resolved audit item: local matter-core boundary (pooled AUC at the 0.85 promotion gate; 5/8 regimes strictly above). The chirality-flip is established at the global coefficient level: the bounded-operator readouts of $\theta_{\text{chir}}(N)$ on the lattice N-ladder transition through $\pi/4$ at $N_{\text{inv}} = d\cdot N_{\text{gen}}\cdot N_* = 600$ under the first-principles running $\tan\theta_{\text{chir}}(N) = N_{\text{gen}}^{2x-1}$, $x = \ln(N/N_*)/\ln(d\,N_{\text{gen}})$. A stronger claim — that $\theta_{\text{chir}}(a) > \pi/4$ at lattice node a corresponds to local matter-core support $T_{00}(a) > 0$ and $a \in C_N(\tau)$ — has been audited on per-node lattice data under two distinct definitions. A first amplitude-level definition takes $\theta_{\text{chir}}(a) = \arctan_2\big(|\gamma(a)|, |\alpha_\xi(a)|\big)$ where $\alpha_\xi(a), \gamma(a)$ are the per-node row-means of the chirality-coupling factor fields, and tests (i) AUC of $\theta_{\text{chir}}(a)$ as predictor of $a \in C_N(\tau)$; (ii) Spearman correlation of $\theta_{\text{chir}}(a)$ with $T_{00}(a)$; (iii) Spearman correlation of $\theta_{\text{chir}}(a)$ with the per-node curvature-time-time deviation $|R_{00}(a)| = \big||\psi(a)|^2 - \langle|\psi|^2\rangle\big|$ where the local order parameter ψ is bundled, falling back to the analogous deviation on the Ξ field where ψ is not bundled; (iv) conditional-probability ratio $P(a \in C_N | \theta > \pi/4)/P(a \in C_N | \theta \le \pi/4)$; (v) Spearman correlation with the signed distance to ∂C_N. The promotion gate was declared up front as AUC ≥ 0.85 *and* $\rho(\theta, T_{00}) \ge 0.5$.

Result on 13 regimes spanning the P_3–P_6 ladder at small N plus the P_5 ladder at $N = 64, 100$ (2672 lattice nodes total): pooled AUC $= 0.506$ (random-classification, far below the 0.85 gate); Spearman $\rho(\theta, T_{00}) = +0.148$ (well below the 0.5 gate); $\rho(\theta, |R_{00}|) = +0.022$; $\rho(\theta, \text{dist}) = +0.009$. Critically, 0/2672 lattice nodes have local $\theta_{\text{chir}}(a) > \pi/4$ at all: the per-node $\arctan(|\gamma(a)|/|\alpha_\xi(a)|)$ sits below $\pi/4$ on every site of every seed of every regime tested ($|\alpha_\xi(a)| > |\gamma(a)|$ uniformly), so the threshold $\pi/4$ does not partition the lattice into a matter-side and a vacuum-side at the node level. The conditional-probability ratio in (iv) is therefore structurally undefined under the simple amplitude-level promotion — the diagnostic outcome of the test.

A second audit replaces the amplitude-level promotion with a local RG-window definition that lifts the global running formula site by site, $\theta_{\rm chir}(a) = \arctan\!\big(N_{\rm gen}^{2x_{\rm loc}(a)-1}\big)$ with $x_{\rm loc}(a) = \ln(N_{\rm eff}(a)/N_*)/\ln(d\,N_{\rm gen})$ and a per-node effective lattice scale $N_{\rm eff}(a)$. The extended-scope audit covers 13,856 lattice nodes across 22 regimes spanning $N = 36$–300 and including the post-flip regimes $N = 200, 256, 300$ where $\theta_{\rm chir}(N) > \pi/4$ globally; the matter-core support C_N is taken in three independent forms: top-decile of the per-node energy density $t_{00}(a)$, top-decile of the per-node closure residual $\Delta(a) = \|R^{(\rm ED)}(a)\|/\|T(a)\|$, and the per-node topological winding number $|w(a)| > 1/2$. Six $N_{\rm eff}$ candidates were tested; the empirical winner is the row-variance of the Ξ-displacement to the lattice neighbours,

$$N_{\rm eff}(a) \;=\; N\,\frac{{\rm Var}_j(\Xi_{a,j})}{{\rm med}_a{\rm Var}_j(\Xi_{a,j})},$$

selected independently by its Spearman rank correlation $\rho(N_{\rm eff}, \Delta) = +0.54$ on a representative seed.

The threshold $\pi/4$ is the global flip asymptote of the running formula and reduces the per-node test to its matter-regime limit; the proper per-regime matter-side classifier is the regime-relative offset $\Delta\theta(a) \equiv \theta_{\rm chir}(a) - \theta_{\rm glob}(N_{\rm regime}) > 0$. With this physically correct threshold, pooled metrics over the 10,976 lattice nodes across the eight ψ-bundled regimes on the $P_5/P_6/P_8$ ladder are AUC $= 0.849$ $[0.838, 0.860]$ and matter-core enrichment ratio $P(C_N|\Delta\theta > 0)/P(C_N|\Delta\theta \le 0) = 11.31\times$ $[9.24, 14.19]$ (95% bootstrap CIs, 1000 resamples) against $t_{00}(a)$; AUC $= 0.616$ $[0.594, 0.637]$ and ratio $1.75\times$ $[1.57, 1.97]$ against $\Delta(a)$. Per-regime AUC against $t_{00}(a)$ ranges 0.83–0.90 across all eight regimes (peak 0.904 at $N = 64$) with regime-relative ratios 9.4–27×, exceeding the strict 0.85 promotion gate within five of eight regimes and at the gate when pooled. Leave-one-regime-out: the row-variance of Ξ is the top per-node candidate in 8/8 folds for both the $t_{00}(a)$ and $\Delta(a)$ targets, with held-out test AUC 0.83–0.90. Within-regime permutation null (1000 replicates): observed AUC is $z = 37.9\sigma$ above the null mean for $t_{00}(a)$ and $z = 12.7\sigma$ for $\Delta(a)$, $p \le 0.001$. The chirality- flip is therefore both a global coefficient regime change AND a strong local matter-core indicator at the per-node level: $\Delta\theta(a) > 0$ is a strong local predictor of high $t_{00}(a)$ matter loading. The global chirality-flip identity — running rate $21.38°/\ln N$ (0.38% match), $N_{\rm inv} = 600$ (1.5% match), refined β_π (0.6% mean match), all-or-nothing per-regime flip pattern at $N \sim 173$ — is independently confirmed by this audit.

The cosmological-anchor table (Section 8) closes the loop: every Einstein-side observable in the bridge has its prediction independently anchored to its canonical Planck 2018, PDG, or NuFIT 6.0/6.1 measurement, and the 16-row anchored table verifies 12 EXACT, 2 PRECISE and 2 ORDER landings (plus 2 structural-tier rows) with maximum EXACT/PRECISE residual 1.385%. The corpus, taken as a whole, is therefore not a proof of consistency in the abstract but a quantitative, parameter-free match against the standard-model and cosmological measurements that any candidate UV theory must reproduce.

Version-lock matrix. The cross-paper consistency claims of the present note are anchored to the specific manuscript versions of the companion papers shipped in this bundle. The SHA-256 content hashes of each manuscript file (auto-generated by `src/make_version_lock_table_tex.py` at the moment this note is compiled) are tabulated below for forward reproducibility: any reader can recompute the SHA-256 of each `paper/manuscript.tex` (or `bridge_note.tex`) to verify that the version they hold matches the version under which the present consistency note was written.

Paper	Repo	SHA-256 (12 hex)	Accepted claim level	Depends on
P1	`hbr-ew-scale`	`8872b085f7d1`	conditional EW closure	S_{bounce}, T-parity, MS-bar
P2	`causal-wave-landings`	`b91aad0f41bd`	$\mathcal{R}$-coefficient backbone (hypothesis)	P3 library, P4 metric, T-parity
P3	`loop-class-closure`	`98ffbd422bc4`	loop-class library + topology-tuple protocol	$\mathcal{R}$ from P2
P4	`egr-closure`	`8843d51299a2`	bulk-percentile Einstein-class closure	Λ^{back}, P2 $\mathcal{R}$
P4-A	`egr-schwarzschild-ppn`	`cbc2b3771d3b`	linearised Schwarzschild + PPN	P4 closure
P4-B	`egr-anisotropic-source-dm-de`	`e6a3914e8590`	source decomposition + halo audit	P4 closure, P3 strict D_{Ω}
P4-C	`egr-h3c-witnesses`	`b64349c1cf91`	indirect-witness chain	P4 closure
P4-D	`egr-atiyah-singer-chirality`	`73c4bc1ad24e`	native/trivial Wilson-Dirac index audit + non-trivial Kogut–Susskind staggered flux closure; physical chirality lift conditional	P4-C chirality witness
P5	`qft-einstein-bridge`	`45b61d07dc1c`	technical consistency note (not a UV completion; cf. P6)	P1, P2, P3, P4 versions hashed below
P6	`relational-uv-closure`	`(see P6 repo)`	UV-closure construction (this is the UV-completion companion)	P1, P2, P3, P4, P4-A, P4-B, P4-D

Table 2: Version-lock matrix for the P1–P5 collection. Each row gives the manuscript-file SHA-256 (truncated to 12 hex digits for compactness; full hash recoverable by sha256sum on the file), the accepted claim level (what each paper does and does not assert), and the inter-paper dependencies the bridge note relies on. The present technical consistency note inherits the limitations of every paper it depends on.

10 Limitations

- *Bridge scope.* This is a structural bridge, not a unification proof. The bridge claim is that the electroweak and gravitational / cosmological sectors share the same finite loop-factor library, the same coefficient-reduction system $\mathcal{R}$, the same shared resummation kernel $1/(1-2\gamma^2)$, and the same emergent time-orientation; it does not derive a UV completion of either sector. The full electroweak-scale closure is in the companion electroweak-scale paper [2]; the full gravitational-closure (Schwarzschild defect, PPN, BH sector) is in [5]; and the UV-closure construction over the same two-phase carrier — UV state space, unified UV action, two-phase coefficient theorem, sector-by-sector self-adjointness, and the closure ledger under the four-tier convention closed-by-theorem / closed-by-audit / closed-conditional / open-target — is in the companion UV-closure paper [9].

- *Independent numerical predictions.* The bridge does not introduce numerical predictions independent of those in the four cited preprints [2–5]; the 16-row anchored (+2 structural-tier) Einstein-sector cosmological-anchor table of Section 8 and the readouts collected in Section 9 are derived from the same five coefficients and finite loop-class library as the cited preprints, evaluated here against Planck 2018, PDG, and NuFIT 6.0/6.1 [16] anchors. The bundled `src/check_bridge_consistency.py` verifies that the loop-class library, the $\mathcal{R}$-reduction values, and the four shared conditionals reproduce identically across sector assignments.

- *Topology-tuple reading.* The spinor-trace component count $n \in \{0, 1, 2, 4\}$ is read off uniquely from each observable's tree-level Dirac structure (Theorem 1 of [4], Section 2); $n = 0$ is reserved for two-loop compounds carrying no active spinor line. The mapping is single-valued and the reproducibility package implements it through a JSON-bundled (n, g, s, w, r) tuple per observable. A general automated extractor that ingests an arbitrary Lagrangian and emits the tuple without per-observable bundling is a tooling task left for future work.

- *Sector discrimination.* The component count n does not by itself sort observables into electroweak vs. gravitational classes: the QFT and Einstein sides both contain $n = 1$ entries, and the discrimination is made at the level of physical sector (Section 5).

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Code and data availability

Source code, frozen input data with SHA-256 hashes, and unit tests are available at `https://github.com/TerraSignum/qft-einstein-bridge-repro` under the MIT license. The published results correspond to release `v0.1.0`, archived on Zenodo with DOI `10.5281/zenodo.XXXXXXX`.

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